Shortest path algorithm

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Solution of the practice quiz

Before we apply the algorithm, we note that all costs in the network are non negative. Therefore, there is no cycle with negative costs, and the problem is bounded.

Each iteration of the algorithm is summarized in Table 1, where

- the first column reports the iteration number,
- the second column reports the content of set $\mathcal{S}$,
- the third column reports the node that is selected to be treated in the next iteration,
- all the other columns contain the labels of all the nodes in the network.

We comment below the first iterations.

Iteration 0 We initialize the set $\mathcal{S}$ with the starting node 1. All labels are set to ∞, except $\lambda_1 = 0$. In column i, we select the node that will be treated by the next iteration. Here, there is only one node: $i = 1$.

Iteration 1 We treat node 1.

- We remove node 1 from $\mathcal{S}$.
- We consider all arcs leaving node 1:
  - Arc (1, 2): the candidate label for node 2 is $\lambda_i + c_{ij} = 0+8 = 8$. As it is inferior to the current label (∞), we update $\lambda_2 = 8$ and include node 2 in set $\mathcal{S}$.
  - Arc (1, 3): the candidate label for node 3 is $\lambda_i + c_{ij} = 0+14 = 14$. As it is inferior to the current label (∞), we update $\lambda_3 = 14$ and include node 3 in set $\mathcal{S}$.

Iteration 2 We select the node that will be treated by this iteration. We have the choice between 2 and 3. We arbitrarily choose to treat node 3.

- We remove node 3 from $\mathcal{S}$.
- We consider all arcs leaving node 3:
  - Arc (3, 5): the candidate label for node 5 is $\lambda_i + c_{ij} = 14 + 11 = 25$. As it is inferior to the current label (∞), we update $\lambda_5 = 25$ and include node 5 in set $\mathcal{S}$.

Iteration 3 We select the node that will be treated by this iteration. We have the choice between 2 and 5. We arbitrarily choose to treat node 2.

- We remove node 2 from $\mathcal{S}$.
- We consider all arcs leaving node 2:
  - Arc (2, 3): the candidate label for node 3 is $\lambda_i + c_{ij} = 8 + 4 = 12$. As it is inferior to the current label (14), we update $\lambda_3 = 12$ and include node 3 in set $\mathcal{S}$.
  - Arc (2, 4): the candidate label for node 4 is $\lambda_i + c_{ij} = 8 + 9 = 17$. As it is inferior to the current label (∞), we update $\lambda_4 = 17$ and include node 4 in set $\mathcal{S}$.
  - Arc (2, 5): the candidate label for node 5 is $\lambda_i + c_{ij} = 8 + 7 = 15$. As it is inferior to the current label (25), we update $\lambda_5 = 15$. We do not need to include node 5 in set $\mathcal{S}$, as it already belongs to it.

Iteration 4 We select the node that will be treated by this iteration. We have the choice between 3, 4 and 5. We arbitrarily choose to treat node 3.

- We remove node 3 from $\mathcal{S}$.
- We consider all arcs leaving node 3:
  - Arc (3,5): the candidate label for node 5 is $\lambda_i + c_{ij} = 12 + 11 = 23$. As it is not inferior to the current label (15), we do not do anything.

Iteration	S	i	λ_1	λ_2	λ_3	λ_4	λ_5	λ_6	λ_7	λ_8	λ_9
0	$\{1\}$	1	0	∞	∞	∞	∞	∞	∞	∞	∞
1	$\{2,3\}$	3	0	8	14	∞	∞	∞	∞	∞	∞
2	$\{2,5\}$	2	0	8	14	∞	25	∞	∞	∞	∞
3	$\{3,4,5\}$	3	0	8	12	17	15	∞	∞	∞	∞
4	$\{4,5\}$	4	0	8	12	17	15	∞	∞	∞	∞
5	$\{5,6\}$	5	0	8	12	17	15	34	∞	∞	∞
6	$\{6,9\}$	6	0	8	12	17	15	18	∞	∞	23
7	$\{7,8,9\}$	7	0	8	12	17	15	18	23	27	23
8	$\{8,9\}$	8	0	8	12	17	15	18	23	26	23
9	$\{9\}$	9	0	8	12	17	15	18	23	26	23
10	$\{\}$	-	0	8	12	17	15	18	23	26	23

Table 1: Iterations of the shortest path algorithm

The other iterations proceed in the same way, until set $\mathcal{S}$ is empty.

To identify the shortest paths tree, we calculate $\lambda_i + c_{ij}$ and λ_j for each arc, where the labels λ are those obtained at the end of the algorithm.

$$
\begin{array}{cccl}
(i,j) & \lambda_j & & \lambda_i + c_{ij} \\
\hline
(1,2) & 8 & = & 0+8, \\
(1,3) & 12 & < & 0+14, \\
(2,3) & 12 & = & 8+4, \\
(2,4) & 17 & = & 8+9, \\
(2,5) & 15 & = & 8+7, \\
(3,5) & 15 & < & 12+11, \\
(4,5) & 15 & < & 17+12, \\
(4,6) & 18 & < & 17+17, \\
(5,6) & 18 & = & 15+3, \\
(5,9) & 23 & = & 15+8, \\
(6,7) & 23 & = & 18+5, \\
(6,8) & 26 & < & 18+9, \\
(7,8) & 26 & = & 23+3, \\
(9,8) & 26 & < & 23+9.
\end{array}
$$

The shortest path tree is composed of the arcs for which the two quantities are equal, as represented in Figure 1.

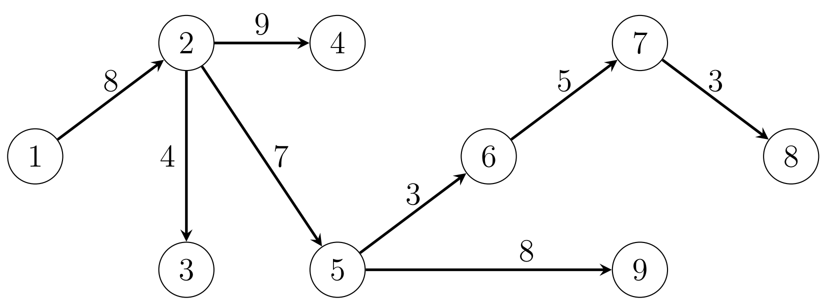

Figure 1: Shortest path tree